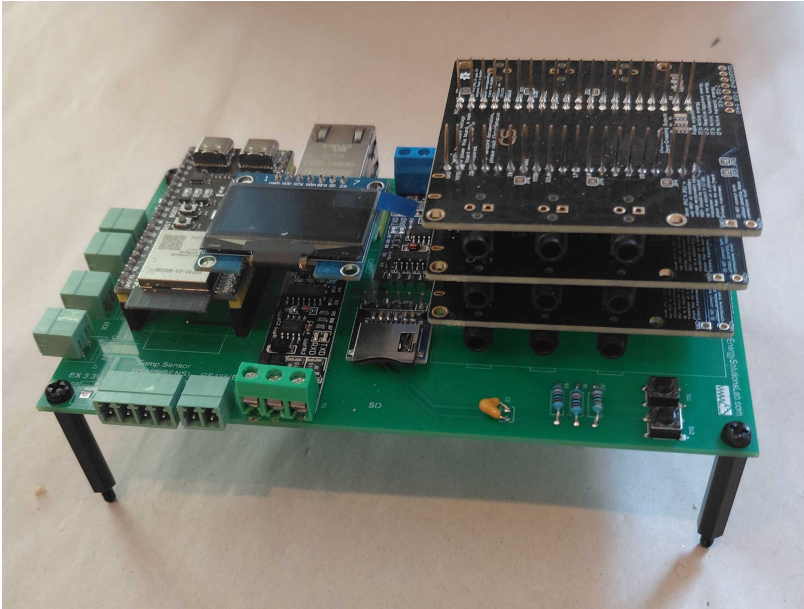


STREET-EMS

Front-of-Meter Feeder Control for the Village Minigrid Street Cabinet



Designing a village minigrid? **STREET-EMS** is the open-source feeder controller for the integration engineer building the **street pedestal cabinet** — coordinating breakers and contactors, metering each branch, and balancing distributed generation across the low-voltage feeder. Front-of-meter, peer-to-peer, and autonomous at the edge.

What Sets STREET-EMS Apart:

- ▶ **Cabinet-Ready Control** : drives contactors and breaker coordination from an onboard relay plus an 8-channel I²C SSR expansion bank for per-branch switching.
- ▶ **Open & Vendor-Agnostic** : built on the open-source meshEMS stack; integrates existing meters and DERs over Modbus, with no proprietary lock-in.
- ▶ **Edge-Native Balancing** : feeder metering and load balancing run locally on a dual-core edge MCU, with no dependence on cloud or central SCADA.
- ▶ **Open-Source Hardware** : released under CERN-OHL-S v2 — schematics, board files and firmware published on GitHub (STREET-EMS).
- ▶ **Field-Serviceable** : built from open, off-the-shelf commodity modules, snap-in mounted for fast in-country assembly and repair.

From a single street pedestal, STREET-EMS meters every branch, switches loads via contactors, and coordinates DERs — bringing intelligent, balanced power to the village minigrid feeder.

Intelligent

- ▶ Multi-branch metrology: V, I, P, Q, PF, import/export kWh
- ▶ Per-channel kW / kWh trip rules for load shedding
- ▶ Roadmap: SIP energy-session admission

Interoperable

- ▶ 2× RS-485 (Modbus) + CAN 2.0
- ▶ 8-ch I²C SSR for contactors / breakers
- ▶ WiFi / BLE & wired Ethernet

Resilient

- ▶ Full edge autonomy without backhaul
- ▶ Onboard + expansion relay control
- ▶ microSD logging | OLED status

PRODUCT	
Product name	STREET-EMS (10Power X-EMS, supplied by NESL)
Model / PCB	NESL EMS Controller PCB 886B
Firmware	meshEMS / OpenAMI metering (open source, ~50k LOC middleware)
License & source	Hardware: CERN-OHL-S v2 Repository: GitHub, STREET-EMS
Serviceability	Open, off-the-shelf commodity modules snap-in mounted for fast in-country assembly and repair
COMPUTE	
Processor	Dual-core 32-bit MCU @ 240 MHz WiFi + Bluetooth LE
Memory	16 MB Flash + 8 MB PSRAM
ASSET COMMUNICATION	
Serial buses	2× RS-485 Modbus: EVC + DTM/sensor ports 1× CAN 2.0
On-board metering	SPI poly-phase metering, up to 6 CT channels V, I, P, Q, PF, freq
Local I/O	2× onboard SSR door-switch input 2 analog buttons
SUPPORTED MODULES & SENSORS (out of the box)	
Energy meters (Modbus)	Single-phase Modbus energy meters import/export kWh
Relay expansion	8-channel I ² C solid-state relay (SSR) bank per-channel kW / kWh trip rules
Environment sensor	Temperature / humidity sensor (Modbus)
DER interoperability	SunSpec models 1 / 11 / 213 CAN 2.0
NETWORKING	
LAN	10/100 Ethernet RJ45
Wireless	WiFi 802.11 b/g/n Bluetooth 5 LE
DISPLAY & STORAGE	
Display & storage	1.3 in OLED display microSD card slot
POWER & ENCLOSURE	
Operating voltage	5 V DC — USB-C or screw terminals <i>[to confirm]</i>
Power / dimensions	<i>[to confirm]</i>
STANDARDS & POSITIONING	
Grid position	Front-of-meter (FTM) LV feeder street pedestal cabinet
Target standards	IEEE ISV (front-of-meter feeder) OpenAMI metrology
Capabilities	Peer-to-peer DER & feeder balancing contactor / breaker switching

Customizable with added hardware modules and open or licensed code stacks. Specifications are preliminary; items marked *[to confirm]* are to be confirmed on the production unit. STREET-EMS is an open-source X-EMS product supplied by New Energy Solutions Lab for 10Power.